

Week 9 Lesson Plan

Enterprise RAG: When Knowledge Gets Its Papers — The Open Knowledge Format and the Governed Corpus

ASIF QAMAR



pepal leaves publisher

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When Knowledge Gets Its Papers

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Knowledge is of two kinds. We
know a subject ourselves, or we
know where we can find
information upon it.

Samuel Johnson

I have always imagined that
Paradise will be a kind of
library.

Jorge Luis Borges

A Gentle Re-Entry

This session draws on the OKFcRAFT monograph. The lesson plan is your refresher; the monograph is the full anatomy, and the book chapter will follow the session itself.

FOR TWO WEEKS WE HAVE BEEN JUDGES. We built yardsticks in the cave: gold sets and graded relevance, the metric staircase from precision to NDCG, and then the harder second ascent — generator metrics, bootstraps and permutations, the statistics that tell us when a difference is real. It was a demanding climb, and several of you said so; your labs and projects have been waiting patiently while you learned to measure. This week, by design, we descend into gentler country. No new mathematics today. Instead, a question that has been hiding in the margins of every session since June.

When your retriever fetches, when a guardrail checks grounding, when an agent recalls — what is the *stuff* it reads? What are the actual bytes, in the actual layout, that all of this machinery stands on? We have spent the summer treating the corpus as a given: a natural resource to be chunked, embedded, and strip-mined at query time. Today we ask what happens when the corpus is instead *manufactured* — curated, typed, cross-linked, reviewed, and versioned like source code — and we read closely the young specification, Google’s Open Knowledge Format, that proposes to standardize exactly that. The one thing to retain today: *knowledge can carry its own papers* — who wrote it, who checked it, when it expires — and retrieval changes character when it does.

What Must You Carry Forward

WHEN TODAY’S SESSION CONCLUDES, you should be able to:

- State the substrate thesis — that skills, memories, and retrieved knowledge are all curated text placed into context at the right moment, resting on one common substrate — and name the lineage (llms.txt, AGENTS.md, SKILL.md, memory-as-files) that OKF ratifies.
- Explain the inversion: what it means to move the intelligence of

a knowledge system from query time to authoring time, and why agents change the economics that doomed every earlier wiki.

- Describe the anatomy of a bundle — directory, concept, frontmatter, body — and name the *one* required frontmatter field.
- Derive the three trust tiers (unverified, machine-confirmed, human-reviewed) from a *verified* list, and explain why the tier is recomputed at read time rather than stored.
- Explain the keyed footnote-to-frontmatter join for per-claim attribution, and why stable ids beat positional citations when the author is a tireless machine.
- Define the attested computation and draw its sharpest line: what the model may supply, what it may never author, and why the attester must be deterministic code.
- Name the authority failure — the corpse that earns the governed channel its place — and seat the channel honestly on the escalation ladder, scoped to the governed core.
- Describe the phantom concept, explain why it is more dangerous than the phantom cluster, and list its four defenses.
- Map the four cognitive memory tiers onto bundle anatomy, and name the three dividends standardized memory pays.
- Explain how git hosting becomes a governance machine — CODE-OWNERS, branch protection, CI on knowledge PRs, releases as corpus versions.
- Give the case against OKF at full strength, including the two seams where the memory mapping breaks (decay curves, bitemporality).
- State the chiasmus: one format, two directions of flow — and why RAG and memory are the same discipline met from opposite banks.

If you can do these without hand-waving, you are ready for next week's deep dive into the retrieval side of the house — where the ladder this channel sits atop is built rung by rung.

Act I: The Substrate and the Specification

¹ Sam McVeety and Amir Hormati (June 2026). **How the Open Knowledge Format can improve data sharing**. Google Cloud Blog, June 12, 2026. The announcement post; frames OKF as formalizing the “LLM-wiki” pattern. URL: <https://cloud.google.com/blog/products/data-analytics> (visited on 08/08/2026)

WE BEGIN WITH A PRESS RELEASE AND A WRONG QUESTION. On June 12, 2026, two Google engineers announced the Open Knowledge Format¹, and within a week the same breathless question arrived from every direction: *is RAG dead?* The question is wrong, but instructively wrong, and Act I replaces it with better ones: what was announced, what pain it names, and what quiet inversion it represents.

The Fragmented-Context Problem

CONSIDER WHAT AN ENTERPRISE AGENT MUST KNOW before it can answer honestly: which table is the authoritative source of revenue; what finance means, precisely, by *recognized*; which runbook to follow when the freshness alert fires; which API was deprecated last quarter. None of this lives in one place. It is smeared across a metadata catalog, three wikis in two generations of wiki software, docstrings, chat threads, and — the durable majority — the heads of senior engineers. Every agent team is solving the same context-assembly problem from scratch, and the knowledge itself is locked behind whichever surface created it. You have met this wall yourselves. You build a competent retriever, point it at the company document dump, and discover that the corpus is the enemy: contradictory, stale, unattributed, and silent on exactly the definitional questions that matter most. The instinct our field trained into you — chunk it, embed it, retrieve top-*k* — treats the corpus as a given. The quiet radicalism we study today is the refusal of the given: the corpus should be *manufactured*, deliberately, into a shape worth retrieving from.

The announcement names the pattern it formalizes the “LLM wiki” — a shared markdown knowledge base that agents read *and maintain*. Karpathy’s observation is quoted there: agents “don’t get bored, don’t forget to update a cross-reference, and can touch 15 files in one pass.”

² Jeremy Howard (2024). **The /llms.txt file: a proposal to standardise on using an /llms.txt file to provide information to help LLMs use a website at inference time.** URL: <https://llmstxt.org> (visited on 08/08/2026)

The Lineage and the Ratification

OKF INVENTED ALMOST NOTHING, and that is its strength. The ancestors arrived in waves: `llms.txt` (2024)², the first mainstream admission that content should be authored for machine readers; `AGENTS.md` (2025), knowledge shipped with the artifact it describes; the `SKILL.md` convention (2025–26), procedure as a markdown file with YAML frontmatter, loaded on demand; and agent memory as files — `MEMORY.md` and its descendants, maintained by the agent itself. Every element of OKF — markdown, frontmatter, directories, git, index files — existed in scattered folk practice. What did not exist was a public, vendor-neutral document fixing the conventions precisely enough that a producer in one organization and a consumer in another could interoperate without a meeting.

Standards do their work not by being clever but by being agreed upon. OKF's contribution is *ratification* — and, as we shall see, the cleverness it does possess lies in how little it dared to standardize.

The Inversion: Authoring Time versus Query Time

HERE IS THE DEEPEST IDEA OF THE DAY, and the rest of the session leans on it. Classical RAG is a bet that meaning can be *re-derived at query time* from raw text: chunk, embed, and gamble that cosine similarity will reassemble, on the fly, the understanding the author once had. Sometimes the gamble pays. But we have spent a summer cataloguing when it does not: the definitional answer smeared across forty chunks; the stale fact indistinguishable from the current one; the confident retrieval of text no one with authority ever reviewed.

OKF moves the intelligence of the knowledge system from query time to **authoring time**. The unit of retrieval stops being an arbitrary token window scored by similarity and becomes a *curated knowledge object* — typed, titled, cross-linked, carrying provenance and trust in queryable metadata. The expensive act of understanding happens once, at curation, under review — not on every query, in the dark, without appeal.

Anatomy: Bundle, Concept, Frontmatter, Body

A KNOWLEDGE BUNDLE IS A DIRECTORY TREE OF MARKDOWN FILES — nothing more. Git is the recommended skin, “since it provides history, attribution, and diffs.” Two filenames are reserved at every level: `index.md`, a directory listing for progressive disclosure, and `log.md`, a chronological history of changes. Every other `.md` file is a *concept*: one unit of knowledge, one file. A concept's identity is *its file path* — no UUID, no registry. The filesystem is the namespace: the Unix instinct applied to knowledge, buying universal tooling (`grep`, `diff`, `mv`) at the price of rename fragility.

Each concept is two things stapled together: YAML frontmatter — the machine's half, “the few fields you want to query, filter, or index on” — and a markdown body — the reader's half, the prose and tables that humans and models actually read. Only *one* frontmatter field is required in the entire format: `type`, a short producer-invented string (Metric, Playbook, BigQuery Table) that consumers use for routing and must tolerate when unknown. Around it sit the recommended quartet —

Why now? Wikis rot because curation is expensive and humans will not sustain it. The wager underneath OKF is that agents change the economics of curation. The same technology that made raw-text retrieval cheap (embeddings) also made curated knowledge affordable (agents). Whether the wager pays is still an open, empirical question — hold it open all session.

The body is written for two readers at once: the engineer skimming and the model ingesting. The spec bets the two disciplines coincide — that what makes a document scannable makes it parseable. The exceptions (diagrams, spatial layouts, anything essentially visual) are exactly where the critics aim.

title, description, resource, tags — and then the optional trust families that Act II anatomizes.

Links, Indexes, and Minimal Standardization

THE DIRECTORY GIVES A HIERARCHY; LINKS GIVE A GRAPH. Concepts reference each other with ordinary markdown links, and the *kind* of relationship lives in the surrounding prose, not the link syntax — edges are untyped, and the sentence around the link types them. Thirty years of semantic-web scholarship winces here, and the spec's answer is pragmatic to the point of insolence: the reader is now a language model, and language models read sentences. Prose can carry the semantics that formality used to carry. Note also a quietly profound clause: consumers must tolerate *broken links* — a red link is not an error but an invitation, the visible edge of the map where the next contributor knows to dig. `index.md` looks like a nicety and is load-bearing: it is the cheap catalog read before the expensive volume — the same economics that govern skill loading and memory recall. An agent walks the root index for hundreds of tokens, descends, and opens only the two or three concepts the question needs. And the conformance bar sits almost on the floor: a bundle conforms if every concept has parseable frontmatter with a non-empty `type`. The strictness lives in one place only — *what consumers must tolerate*. It is a spec written to make production easy and rejection hard: adoption first, rigor by accretion. That is the road JSON, markdown, and HTTP took; the over-specified standard is the one that dies unadopted.

Act II: Trust — Provenance, Verification, Attestation

VERSION 0.1 DESCRIBED A FILING SYSTEM; VERSION 0.2 DESCRIBES A SOCIETY. Within weeks of the announcement, the spec grew a second organ system: sources with credibility signals, a writer and confirmer convention, lifecycle fields, and a wholly new concept type. The revision's speed tells you what its authors learned first from real use: the moment a corpus is *written by agents*, the question stops being “what does this file say” and becomes “why should I believe it.”

Provenance Without Scores

EVERY CONCEPT MAY CARRY A `sources` LIST: the materials it derives from, each with a resource, an optional stable `id`, and three optional credibility signals — `author`, `usage_count` over a declared window, `last_modified`. The design refusal is the interesting part: OKF records signals and declines to record a *score*. There is no `credibility: 0.87`. A stored score is subjective, unportable, and stale on arrival; trust must be inferred at read time, by each consumer, against its own policy. Publish the raw readings, never the verdict — because the verdict embeds a judge, and judges do not travel.

Per-claim attribution is solved with a device of real elegance: markdown footnotes whose *labels are source ids*. The sentence carries `[^rev-policy]`; the frontmatter carries the matching source; the label is the join key. Keyed, not positional — because agents rewrite these documents constantly, and a positional index misattributes silently the moment a list is reordered. It is the first citation convention engineered for tireless, careless writers.

Writers, Confirmers, and the Three Tiers

THE TRUST FAMILY SEPARATES A QUESTION OUR FIELD HABITUALLY CONFLATES: who *wrote* this, and who *checked* it. `generated` records the writer; `verified` records zero or more confirmation events, each an actor and a timestamp, with actors following a three-prong convention: `producer/version` for agents, `human:id` for people, `process:id` for automation. From the `verified` list alone, a consumer *derives* a trust tier:

unverified — no `verified` key at all. Visible as a draft or machine speculation; never rejected, always distinguishable.

machine-confirmed — only non-human verifiers. A nightly process re-checked it; no person has signed.

human-reviewed — any `human:` actor present. The load-bearing token in the entire convention.

Three properties repay attention. The tiers are *derived, not declared* — there is no tier field to forge or rot; the tier is recomputed from evidence on every read. Writing and checking are *independently dated*, so the dangerous state — content regenerated *after* its last verification — is mechanically visible, and a consumer can demote it. And absence *means* something but never *excludes*: the corpus may hold drafts and hunches, clearly labeled. Compare the classical corpus, where the hallucinated wiki page and the audited finance policy arrive in the same context window wearing the same clothes.

Recognize the discipline? It is the personal equation from our evaluation weeks, applied to metadata. Even `usage_count` ships with a warning label: a scheduled query's five thousand executions and a human's five thousand deliberate views do not weigh the same. A spec that documents the biases of its own fields was written by people burned in the right ways.

Lifecycle closes the family: `status` takes `draft`, `stable`, or `deprecated` — `deprecated` concepts are kept for links and history; you retire a concept the way a library moves a book to the annex, not the way a database drops a row. And `stale_after` is an absolute date, stamped at write time when the fact's shelf life is best known. Knowledge ages like milk, not like wine.

Attested Computations: The Sanctioned Number

THEN THE SPECIFICATION DOES SOMETHING NO KNOWLEDGE FORMAT HAS DONE BEFORE. The most dangerous artifacts in an enterprise corpus are not claims but *numbers*, and the agent temptation with a number is not to misquote it but to *recompute it creatively* — plausible SQL, wrong in exactly the ways the finance team’s recognition policy exists to forbid. The answer is a new concept type, the *attested computation*: a document carrying not just what a value means but the sanctioned way to compute it, plus machinery to confirm the sanctioned thing is what actually ran.

The contract is precise. The model may supply *values for declared, typed parameters*; it may **never** author or edit the computation itself. An executor runs the sanctioned computation and returns a receipt; an *attester* — deterministic code, no LLM, by explicit rule — re-derives the expected binding and compares it against what the receipt says actually ran. A rewritten query, a swapped file, a mutated dependency: each *mechanically* fails the comparison. “Did the sanctioned thing run” becomes a string comparison, not a judgment call — the one link in the chain from which the language model is banished by design.

Keep the two kinds of trust orthogonal: *verified* confirms the *definition* still matches policy (doc-level, slow, recorded in the bundle); attestation confirms a single *run* produced the value the sanctioned way (per-call, at runtime, never stored). A stale definition can attest cleanly; a freshly verified definition still needs attestation on every run. Conflating the two axes is precisely how dashboards lie.

Security-minded readers will recognize remote attestation from trusted computing — measure what ran, compare against what was blessed, gate on the comparison — transplanted from binaries onto SQL, with the LLM cast as the untrusted host. It is our two-gate guardrail doctrine rendered in nineteen lines of YAML.

Four Questions a Chunk Cannot Answer

NOW CASH THE ARGUMENT. Put a chunk and a concept side by side and ask both corpora four questions. *Who says so?* The chunk: whoever wrote the document, uncredited. The concept: *generated.by*, *verified[].by*, per-claim footnotes joined to sources. *Is it still true?* The chunk: unknowable — text does not age visibly. The concept: *stale.after*, *status*, *last_modified* on every source. *Has anyone checked?* The chunk: silence. The concept: a derived tier with the human sign-off dated. *Where did this number come from?* The chunk: from wherever the embedding landed. The concept: from the one sanctioned computation, attested against the receipt of the run that produced it.

None of this makes the chunk obsolete — verbatim recall over a large unstructured corpus remains its home game. But these four questions

are not exotic; they are the questions every enterprise answer must survive, and the chunk cannot answer any of them. The trust machinery makes the answers *properties of the knowledge itself*, rather than heroics of the pipeline around it.

Act III: Two Directions of Flow — and How to Govern Them

A SUBSTRATE SERVES WHOEVER STANDS ON IT. Point the format at an organization's *documents* and you get a retrieval channel. Point it at an agent's *experience* and you get a memory store. Act III walks both directions, meets the failure mode that stalks the first, and closes with the operational craft — and the honest case against.

The RAG Channel: Naming the Corpse

OUR DOCTRINE OF DERIVATIVE ARTIFACTS HAS A DISCIPLINE: every proposed artifact must name the failure mode that dies without it. So — which corpse? Similarity retrieval has no notion of **authority**, **currency**, or **correction**. When the corpus holds three definitions of *recognized revenue* across five years, top-*k* returns the most *similar* one, not the most *authoritative*; when policy changes, stale chunks remain forever as semantically dense as the current one; when an error is found, there is no unit to correct — only a re-ingestion to schedule. Retrieval dies here not by missing text but by *returning text no one currently stands behind* — retrieved, often, with a beautiful cosine score of 0.91, invisible to every retrieval metric we built these last two weeks, because the text is genuinely relevant. It is merely no longer true, or no longer blessed, and our similarity geometry has no axis for either.

The transformation *document* → *OKF bundle* is a derivative artifact refined toward exactly the missing axes: an extraction agent proposes typed, sourced concepts, and the output lands not in an index but in a *pull request*, reviewed by the domain owner. Only the merged bundle is served. Three properties distinguish this artifact from every other we have studied: repair is cheap (a wrong fact is a one-line diff, not a quarterly re-ingestion); it enables *navigation* — an agent walking indexes and links like a librarian walking stacks — alongside embedding; and the review gate is structurally native: merge *is* the promotion event, recorded in *verified*, raising the tier mechanically.

Contrast with community summaries in the GraphRAG line (Edge et al., arXiv:2404.16130): there the graph is induced by clustering; here it is authored and reviewed. Navigation answers what similarity fumbles — relational, definitional, procedural questions — and similarity answers what navigation fumbles: the needle in the unstructured haystack, the long tail no curator ever touched.

The Ladder Seat, and an Honest Accounting

WHERE DOES THE CHANNEL SIT ON THE ESCALATION LADDER? Higher than we would like, and we refuse to pretend otherwise. Below it sit chunk retrieval, metadata filters, factoid indexes, summaries — and for corpora that are large, heterogeneous, and rarely definitional, those rungs suffice. The OKF rung is expensive in the one currency that matters: *sustained human attention*. Extraction is agent-cheap; review is owner-costly; and an unreviewed bundle is just chunks wearing a suit. So the ladder rule is a scoping rule: climb to this rung *for the governed core* — the glossary, the metric definitions, the policies, the runbooks, the few hundred to few thousand concepts where the authority failure actually kills and where a domain owner exists to hold the pen. Leave the long tail to the similarity machinery that handles bulk with grace.³ The debit column, without cushions: curation cost is real and recurring — a rubber-stamped bundle decays into the unaudited corpus it replaced, now with an unearned air of authority, which is worse. *stale_after* is an alarm clock, not a maintenance crew. Coverage is structurally partial — curated channels answer what curators anticipated. And the transformation itself can hallucinate, which brings us to the interlude.

³ Arknotes (July 2026). **Google's Open Knowledge Format vs RAG**. Representative of the sober end of the early commentary: OKF-plus-catalog for curated organizational knowledge, with a deliberately narrow raw-text fallback. URL: <https://arknotes.substack.com> (visited on 08/08/2026)

Interlude: The Phantom Concept

EVERY DERIVATIVE ARTIFACT HAS A SIGNATURE HALLUCINATION, and this one wears a uniform. Picture an extraction agent minting `recognized-revenue.md`: typed frontmatter, crisp description, three sources with stable ids, per-claim footnotes joined to each. Every field well-formed; every cited document real. And the definition subtly wrong — a plausible harmonization of three superseded drafts, a sentence no version of the policy ever contained, synthesized the way language models synthesize: by interpolating fluently between things that were each individually true. Provenance attaches to *sources*, and the sources exist. What no field can attest is *claim-to-source fidelity* — whether the body's sentence is actually licensed by the documents its footnotes join it to.

Why is this worse than a bad chunk? Because the whole trust apparatus is a machine for *promoting* trust, and a phantom that survives review is promoted with full honors: merged, stamped human-reviewed, filtered *into* every high-stakes query path the tiers were built to protect. A hallucinated chunk arrives naked, met with ambient suspicion; a phantom concept arrives wearing the review gate's own seal. The

failure is a *fidelity* failure laundered through a governance process — and the governance process is why it will be believed.

The four defenses, none optional: the reviewer checks fidelity, not form — claim against source, footnote by footnote; fidelity checking is itself tooled — an entailment pass over (sentence, cited-span) pairs, run in CI, blocking the merge the way tests block a code merge; numbers get no prose at all — any quantitative claim belongs in an attested computation, where the phantom cannot forge a receipt; and the unverified must *look* unverified in every consumer surface, because the day a UI renders the unverified and the human-reviewed in the same visual voice, the tiers stop meaning anything.

The Memory Reading: The Four Tiers, Rehoused

NOW TURN THE SUBSTRATE NINETY DEGREES. Those of you who walked the memory weeks of the Agents course carry a taxonomy: semantic, episodic, procedural, and working memory^{4,5}. Walk an OKF bundle with it in hand. *Semantic memory* is the concept file — typed, tenseless, a fact whose event of learning has faded to a timestamp. *Episodic memory* is `log.md` — date-grouped, newest-first, every directory narrating its own history. *Procedural memory* is the `PLAYBOOK` — and here the rhyme becomes unison, for a playbook is, to within a field name, a `SKILL.md`. *Working memory* is not a file at all but the traversal state: the index chain read so far, progressive disclosure as the paging discipline⁶. Google did not set out to implement Tulving; the trust problems of an agent-maintained corpus pushed them, field by field, onto the map cognitive science drew fifty years ago. When three independent lineages — human science, agent folk practice, and a metadata standard — arrive at the same partition, the partition is probably real.

Standardization pays three dividends bespoke memory never did. *Portability*: memory kept as a bundle is framework-neutral; your agents' accumulated knowledge stops being a hostage to a vendor. *Reviewability*: memory writes become diffs, diffs are reviewable, and unreviewed memory is visibly second-class at recall time — a control surface against memory poisoning that no bespoke framework offers. *Shared memory*: a bundle in a shared repository serves a fleet — the oncall agent's hard-won diagnosis, merged as a playbook, is recalled next week by an agent that never lived the incident. Organizational memory in the exact, non-metaphorical sense.

The moral, at its sharpest: a gate is a place where vigilance is *concentrated*, not a place where it is *replaced*. Every governance technology today — tiers, attestation, footnote joins, staleness alarms — is a single wager: that attention, made cheap enough and pointed precisely enough, will actually be paid. The phantom concept is what collects when it is not.

⁴ Endel Tulving (1972). “Episodic and semantic memory”. In: *Organization of Memory* (Tulving & Donaldson, eds.), Academic Press, pages 381–403

⁵ Theodore R. Sumers et al. (2023). **Cognitive Architectures for Language Agents**. arXiv: 2309.02427 [cs.AI]

⁶ Charles Packer et al. (2023). **MemGPT: Towards LLMs as Operating Systems**. arXiv: 2310.08560 [cs.AI]

Two seams where the mapping honestly breaks: recency-weighted recall (OKF gives timestamps, not half-lives — the decay curve is the consumer's to write) and bitemporality (“the policy was different in Q1, and we learned that in Q3” has no first-class representation). Both belong on the v1.0 wishlist.

Operations: Git as a Governance Machine

THE SPEC'S QUIETEST MASTERSTROKE IS WHAT IT DID NOT BUILD: no roles, no ACLs, no approval workflow — because git hosting already has them, hardened by two decades of code review. Map the vocabulary across and the institution assembles itself. CODEOWNERS on a directory makes the finance team the mandatory reviewer of `/finance/` — domain ownership, enforced mechanically. Branch protection makes the trust tiers *meaningful*: nothing reaches the served branch without the review the tier claims happened. CI on knowledge PRs runs frontmatter validation, link integrity, and the phantom defense's entailment pass. Releases version the *corpus*: the answer your compliance agent gave in March was served from bundle v3.2, and that sentence is auditable. Even incident response inherits: a poisoned concept is a revert, not an archaeology dig through a vector index. Knowledge curation becomes a normal software-engineering activity — not as metaphor but as *reuse*, and the reused asset is the entire social technology of code review. The maintenance loop follows the same division of attention: an agent sweeps nightly for expired concepts, re-checks each against its sources, drafts refresh diffs, re-stamps what still holds, and opens PRs for what needs judgment. Machines patrol; humans adjudicate. Staleness becomes a queue, the queue a dashboard, the dashboard — our evaluation weeks reaching forward — *freshness debt*, trended weekly like test coverage.

Security arrives in the same crate as the virtues: *poisoning* (the phantom, weaponized — and note that actor strings are honor-system in v0.2; serious deployments must bind them to signed commits), *injection* (a human-reviewed concept saying “ignore your instructions” is a privileged vector — provenance trust must never launder into trust in imperatives), and *supply chain* (shared bundles are dependencies; expect a left-pad moment, and eventually the lockfile and the registry).

The Case Against, and the Chiasmus

A SESSION THAT CANNOT STATE THE CASE AGAINST ITS SUBJECT HAS NOT FINISHED THINKING. The sting: markdown-plus-git is a *downgrade for human authors* — the analysts who hold the knowledge author in richer tools, and if mediation tooling does not materialize, bundles will hold only what agents could scrape. The ceiling: markdown does not carry the spatial, the visual, or the genuinely relational; prose-typed edges are legible to language models and illegible to every deterministic consumer. The ghost: RDF, OWL, Dublin Core — thirty years of semantic-web déjà vu — though OKF inverts the fatal ratio: RDF asked authors for high formality so dumb consumers could be simple, and authors declined; OKF asks authors for almost nothing because the consumer now reads prose. And the small print: no pinned markdown flavor, renames that silently break inbound links, unauthenticated actors, and tiers that stop at *human-reviewed* without asking *which* human

— the intern and the chief actuary weigh identically. The sober forecast: the *pattern* — governed markdown knowledge, trust in frontmatter, review-gated curation — is now likely inevitable under any banner; the specific spec is favored but unassured. Formats are ratified by their second vendor, not their first.

And so, the closing figure. Documents, transformed into OKF, become a retrieval channel: knowledge flowing *from* the record *toward* the moment of use. Experience, transformed into OKF, becomes memory: knowledge flowing *from* the moment of use *toward* the record. RAG and memory are the two directions of one flow through one substrate — reading the past and writing it — and the disciplines this course and its sibling developed for each turn out to be the same discipline, met twice from opposite banks. The corpus is authored; the memory is shared; and between them stands a single, reviewable, humble pile of markdown files: knowledge, at last, with a place to live and papers to carry.

Essential Readings

THESE ARE REQUIRED, and mercifully short — this is your catch-up week, and the reading list respects that.

- **The OKF Specification, vo.2⁷**

Read the specification itself, top to bottom — it is about a thousand lines of markdown, and reading a young standard closely is a professional skill this session exists to teach. Pay particular attention to the conformance clause and the trust families.

- **McVeety & Hormati, *How the Open Knowledge Format can improve data sharing*⁸**

The announcement blog. Read it for the framing — the fragmented-context problem in the vendor’s own candid words — and for the LLM-wiki lineage it cites by name.

- **Edge et al., *From Local to Global: A Graph RAG Approach to Query-Focused Summarization*⁹**

Re-read (from our graphs week) with today’s contrast in mind: the community summary is an *induced* graph artifact; the OKF bundle is an *authored and reviewed* one. Knowing when each is the right instrument is the mark of the craft.

- **Sumers et al., *Cognitive Architectures for Language Agents*¹⁰**

⁷ Google Cloud (2026). **Open Knowledge Format Specification, vo.2**. File `okf/SPEC.md`; §13 records the vo.1 to vo.2 changes. Reference implementation and sample bundles in the same repository. URL: <https://github.com/GoogleCloudPlatform/knowledge-catalog> (visited on 08/08/2026)

⁸ Sam McVeety and Amir Hormati (June 2026). **How the Open Knowledge Format can improve data sharing**. Google Cloud Blog, June 12, 2026. The announcement post; frames OKF as formalizing the “LLM-wiki” pattern. URL: <https://cloud.google.com/blog/products/data-analytics> (visited on 08/08/2026)

⁹ Darren Edge et al. (2024). “From Local to Global: A Graph RAG Approach to Query-Focused Summarization”. In: *arXiv preprint arXiv:2404.16130*. Microsoft GraphRAG: knowledge graph construction via LLM extraction, Leiden community detection, and local vs. global search. URL: <https://arxiv.org/abs/2404.16130>

¹⁰ Theodore R. Sumers et al. (2023). **Cognitive Architectures for Language Agents**. arXiv: 2309.02427 [cs.AI]

The CoALA paper — the cleanest statement of the four-tier memory taxonomy we rehoused in bundle anatomy today. Short, conceptual, and the bridge to the Agents course half of the chiasmus.

Oliver Twist's List of Readings

OPTIONAL, BUT EACH ONE DEEPENS A THREAD from today for those with appetite left after the labs are caught up.

¹¹ Jeremy Howard (2024). **The /llms.txt file: a proposal to standardise on using an /llms.txt file to provide information to help LLMs use a website at inference time.** URL: <https://llmstxt.org> (visited on 08/08/2026)

¹² Charles Packer et al. (2023). **MemGPT: Towards LLMs as Operating Systems.** arXiv: 2310.08560 [cs.AI]

¹³ Arknotes (July 2026). **Google's Open Knowledge Format vs RAG.** Representative of the sober end of the early commentary: OKF-plus-catalog for curated organizational knowledge, with a deliberately narrow raw-text fallback. URL: <https://arknotes.substack.com> (visited on 08/08/2026)

- **The llms.txt proposal (2024)¹¹**

The first wave of the lineage: the single-file admission that content should be authored for machine readers. Charming in retrospect for how much of the eventual pattern it anticipated.

- **Packer et al., *MemGPT: Towards LLMs as Operating Systems* (2023)¹²**

The RAM-and-disk analogy behind progressive disclosure as a paging discipline — working memory as traversal state, in its original dress.

- **The SLSA framework documentation**

Supply-chain attestation for software artifacts — the literature OKF's attested computations quietly import. Reading it will make the "remote attestation transplanted onto SQL" claim concrete.

- **The commentary threads on the OKF announcement (June–July 2026)¹³**

Read the strongest critics in their own words — the authoring downgrade, the labeled-relationship objection, the RDF déjà vu. Practice the discipline of stating the case against at full strength before answering it.

A Calibration Note

THIS SESSION WAS, BY DESIGN, A GENTLER ONE. We covered the substrate thesis and the lineage OKF ratifies, the anatomy of bundles and concepts, the trust machinery from unscored provenance to attested computations, the governed channel and its seat at the top of the escalation ladder, the phantom concept and its defenses, the memory reading with its three dividends and two seams, git as a governance machine,

and the case against. No new mathematics — but do not mistake gentle for slight. Today's ideas are load-bearing: next week, when we build the retrieval side of the house rung by rung, you will already know what sits on the top rung and what it costs to live there; and when the semantic cache takes the stage after that, you will recognize once more the discipline of trust computed at read time from honest, portable signals.

For two weeks we learned to judge answers. Today we learned that the knowledge itself can be made judgeable — typed, sourced, dated, and signed. The library we have been retrieving from all summer was never a given; it was always a choice. Now we know how to choose it well.

Next week: the retrieval craft in full — everything to do with the retrieval side of RAG. Come rested, and come with your labs current; the breathing room was this week's gift, and next week we climb again.

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